PROJECT SYNOPSIS

1. Title of the Project

HawkSight AI:-An Advanced AI/ML Platform for Stock Market Prediction, Strategy Backtesting, and Automated Trading with an Integrated Wallet System

2. Introduction

While algorithmic trading dominates modern financial markets, retail investors remain underserved by fragmented, high-latency, and opaque systems. Existing platforms often provide black-box solutions without offering insights into the predictive drivers, nor do they provide robust tools for validating the real-world performance of trading strategies. This project addresses these gaps by engineering a comprehensive, end-to-end financial analysis and trading platform. We will leverage a sophisticated stack of data science and machine learning libraries to perform in-depth exploratory data analysis (EDA), develop and benchmark multiple predictive models, rigorously backtest trading strategies, and provide model explainability—all within a unified, broker-less ecosystem enabled by an integrated wallet.

3. Objectives

- **To conduct comprehensive Exploratory Data Analysis (EDA)** on historical NSE stock data using Pandas, Matplotlib, and Seaborn to identify statistical properties and potential predictive features.
- **To develop and benchmark a suite of predictive models,** including classical time-series models (Statsmodels ARIMA), gradient-boosted machines (XGBoost/LightGBM), and deep learning networks (TensorFlow/Keras LSTM).
- **To systematically optimize model performance** by employing advanced hyperparameter tuning techniques using the Optuna framework.
- **To implement model explainability (XAI)** using SHAP (SHapley Additive exPlanations) to interpret the decisions of the best-performing model and understand the key drivers behind its predictions.
- **To design and validate trading strategies** based on model outputs using the backtrader framework, evaluating performance on key metrics like Sharpe ratio, maximum drawdown, and annualized return.
- **To deploy the finalized model and strategy** within an interactive web dashboard built with Streamlit for demonstration, and integrate it into a full-stack web application with an integrated wallet.

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4. Problem Statement

Retail trading platforms lack the technical sophistication required for data-driven investing. They fail to provide users with a comparative analysis of different predictive models, offer no tools for rigorous historical backtesting of strategies, and their AI-driven recommendations are often opaque "black boxes." This creates a significant barrier to trust and informed decision-making. There is a need for a unified platform that not only predicts market movements but also validates its strategies, explains its reasoning, and provides a transparent, broker-less environment for execution.

5. Scope of the Project

The project encompasses a full data science workflow, from data acquisition to deployment. The scope includes: sourcing historical data for NIFTY 50 stocks using yfinance; extensive feature engineering to create technical indicators; training and comparing multiple ML/DL models; performing hyperparameter optimization; analyzing model predictions using SHAP; and conducting robust backtesting of signal-based trading strategies. The final output will be a deployable Streamlit application for interactive model analysis and its integration into the broader web platform.

6. Literature Review / Existing System

Our review of existing systems reveals a clear gap in the market. While platforms like Zerodha offer excellent execution, they lack advanced, customizable analytics. Tools like QuantConnect are powerful for backtesting but are complex and still require a broker. Our research indicates a strong academic and industry trend towards:

- **Ensemble and Hybrid Models:** Combining models like XGBoost and LSTMs often yields superior performance.
- **Explainable AI (XAI):** The demand for transparency in financial AI is growing, with frameworks like SHAP becoming industry standard for interpreting complex models.
- **Rigorous Backtesting:** The use of dedicated libraries like backtrader is essential for avoiding common biases (like lookahead bias) and realistically evaluating a strategy's historical performance. Our project is positioned at the intersection of these three cutting-edge domains.

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7. Proposed System / Methodology

Our methodology is a multi-stage data science pipeline designed for rigor and reproducibility.

1. **Data Acquisition & EDA:** Historical daily OHLCV (Open, High, Low, Close, Volume) data for selected NSE stocks will be acquired using yfinance. We will use pandas, matplotlib, and seaborn to conduct EDA, visualize distributions, check for stationarity, and analyze correlations.
2. **Feature Engineering & Preprocessing:** Using scikit-learn and pandas, we will create a rich feature set, including lagging variables, moving averages, RSI, MACD, and other technical indicators. Data will be scaled using MinMaxScaler for neural network compatibility.
3. **Model Development & Benchmarking:** We will establish a baseline performance using a statsmodels ARIMA model. Subsequently, we will develop and train advanced models using XGBoost and a TensorFlow/Keras LSTM network.
4. **Hyperparameter Optimization:** We will leverage Optuna to automate the process of finding the optimal hyperparameters (e.g., learning rate, number of trees, LSTM units) for both the XGBoost and LSTM models to maximize predictive accuracy.
5. **Model Evaluation & Explainability:** The models will be evaluated on a hold-out test set using metrics like Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). The best model will be analyzed with SHAP to generate plots that explain which features are most influential in its predictions.
6. **Strategy Backtesting:** Trading signals generated by the final model will be fed into the backtrader engine. We will simulate the strategy against years of historical data to calculate key performance indicators like Net Profit, Sharpe Ratio, and Maximum Drawdown.
7. **Deployment & Integration:** The trained model, along with SHAP explainers and backtesting results, will be deployed in an interactive Streamlit dashboard. This serves as the core analytical engine to be integrated via API with the main web application.

8. Modules of the Project

1. **Data Ingestion & EDA Module (yfinance, pandas, seaborn)**
2. **Feature Engineering & Preprocessing Module (scikit-learn)**
3. **Model Training & Optimization Pipeline (statsmodels, xgboost, tensorflow, optuna)**
4. **Model Explainability Module (shap)**
5. **Strategy Backtesting Engine (backtrader)**
6. **Interactive AI Dashboard Module (streamlit)**
7. **User & Wallet Management Application (Main Web App)**
8. **API Integration Layer (Flask/FastAPI)**

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9. Expected Outcomes / Deliverables

- A comparative analysis report on the performance of ARIMA, XGBoost, and LSTM models for NSE stock prediction.
- An optimized and trained prediction model with its hyperparameters and evaluation metrics fully documented.
- A set of SHAP visualizations and an accompanying report explaining the predictive factors for the chosen model.
- A comprehensive backtesting report from backtrader, detailing the historical performance of the proposed trading strategy.
- A functional, interactive web dashboard built with Streamlit that allows for loading the model and visualizing its predictions on new data.
- The final integrated web platform with a working demo of the AI-powered auto-trading.

10. Hardware & Software Requirements

- **Hardware:**
  - Development Machine: 16GB+ RAM, Multi-core CPU.
  - Cloud Infrastructure (Optional but recommended): AWS/GCP/Azure VM with GPU access (for faster deep learning model training).
- **Software:**
  - **Environment:** Python 3.9+, Anaconda/Miniconda.
  - **Core Libraries:** numpy, pandas, scikit-learn, matplotlib, seaborn, yfinance.
  - **Modeling Libraries:** tensorflow, xgboost, lightgbm, statsmodels.
  - **Specialized Libraries:** optuna (tuning), shap (explainability), backtrader (backtesting).
  - **Deployment:** streamlit, Flask or FastAPI.

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11. Project Plan / Timeline

Phase	Task	Duration	Key Libraries Used
1	Data Acquisition, EDA & Feature Engineering	3 Weeks	yfinance, pandas, seaborn
2	Model Building (ARIMA, XGBoost, LSTM)	4 Weeks	statsmodels, xgboost, tensorflow
3	Hyperparameter Tuning & Model Selection	3 Weeks	optuna, scikit-learn
4	Model Interpretation & Explainability	2 Weeks	shap, matplotlib
5	Strategy Development & Backtesting	3 Weeks	backtrader
6	Streamlit Dashboard Deployment & API Development	2 Weeks	streamlit, Flask
7	Final Integration, Testing & Documentation	2 Weeks	-
	Total Duration	**19 Weeks**	

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12. Conclusion

This project moves beyond simple stock prediction to deliver a comprehensive, transparent, and rigorously validated trading intelligence system. By integrating advanced modeling techniques with cutting-edge tools for hyperparameter tuning, explainability, and backtesting, we will produce a system that is not only accurate but also trustworthy. The final platform will empower users with a level of analytical sophistication that is currently absent in the retail trading market, setting a new standard for data-driven financial technology.

13. References

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3. Hochreiter, S., & Schmidhuber, J. (1997). "Long Short-Term Memory." *Neural Computation*.
4. *Backtrader Open-Source Project Documentation*.
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